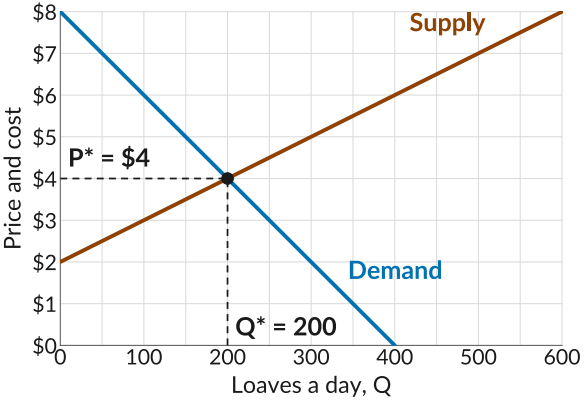
# Who Gets What:
# Allocation, Distribution, and the Gains From Trade

**ECON 201: Principles of Microeconomics**
**Lecture 17**

Div Bhagia

# Last Class: The Bread Market

- With many bakeries selling the same loaf, each is a **price-taker**.
- The supply curve is the marginal cost of the $Q$th loaf, and the demand curve is the $Q$th buyer's willingness to pay.
- They cross at $P^* = \$4$ and $Q^* = 200$ (WTP = MC).

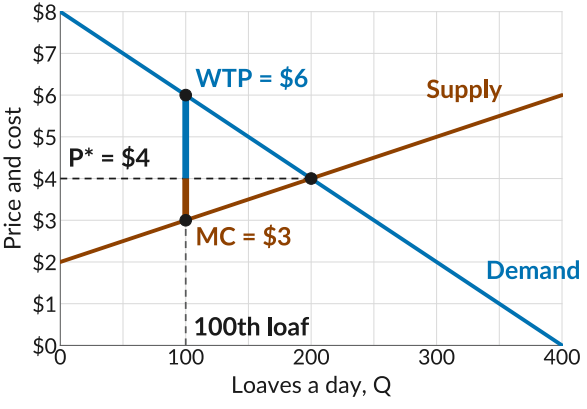



**Today**: who gains from these trades, how the gains are split, and whether the market leaves any gains on the table.

# The Gains From One Loaf

Buyers and bakeries trade because both gain. The **gains from trade** are what each side gets from a trade compared with not trading at all.

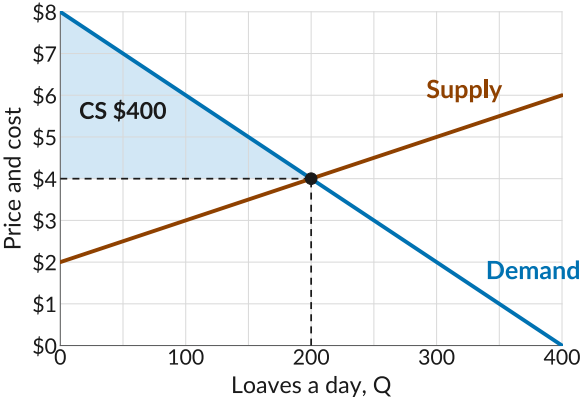
- The 100th buyer would pay up to \$6 but pays \$4, so they gain $6 - 4 = 2$.
- The 100th loaf costs \$3 to bake and sells for \$4, so the bakery gains $4 - 3 = 1$.
- Every buyer who gets a loaf gains WTP $- P$, and every loaf sold earns its bakery $P -$ MC, whoever buys from whom.



# Consumer Surplus

**Consumer surplus (CS)**: the sum of WTP $-$ $P$ over every loaf bought.

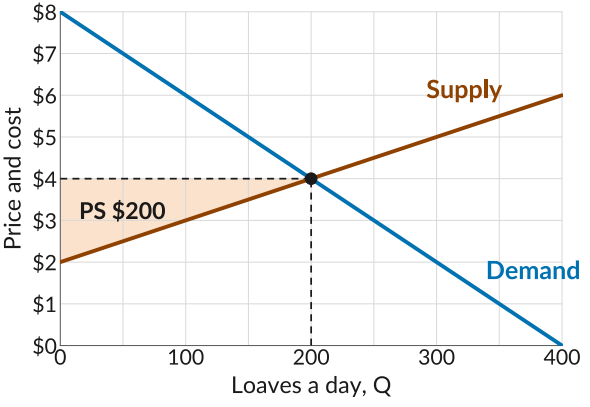
- Every buyer up to the 200th gains the gap between the demand curve and the price.
- Added up:
  $\text{CS} = \frac{1}{2} \times 200 \times (8 - 4) = 400.$



# Producer Surplus

**Producer surplus (PS)**: the sum of $P - \text{MC}$ over every loaf sold.

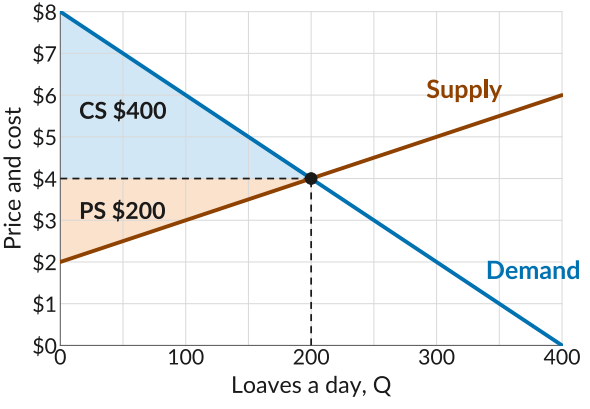
- Every loaf up to the 200th earns its bakery the gap between the price and the supply curve.
- Added up:
  $\text{PS} = \frac{1}{2} \times 200 \times (4 - 2) = 200.$
- Producer surplus is profit plus fixed costs.



# Total Surplus

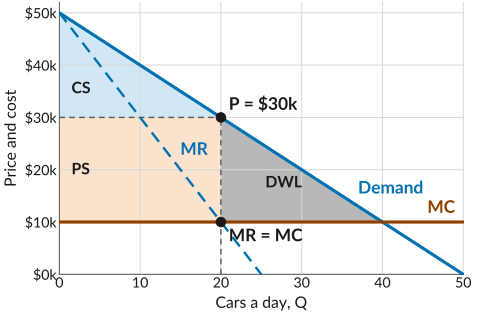
**Total surplus**: consumer surplus plus producer surplus, the gains from all the trades in the market.

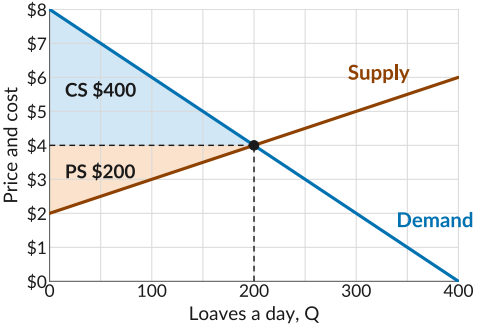
- Total surplus is $400 + 200 = 600$, the whole shaded area.
- Buyers get two thirds of it and bakeries one third.
- Notice any difference from the surplus in the market for Beautiful Cars?



# No Deadweight Loss

Beautiful Cars priced above marginal cost, at the quantity where MR = MC, so buyers who valued a car above its cost went without: a **deadweight loss**.



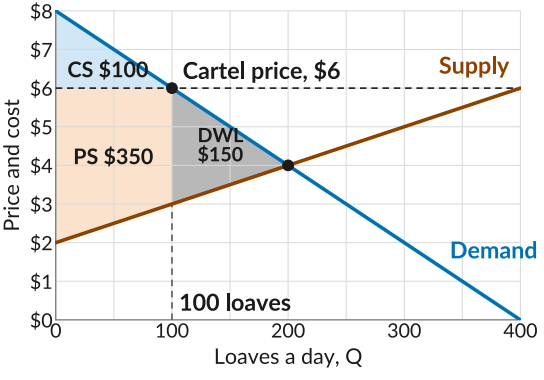




In the bread market, every loaf worth more than it costs gets baked: a competitive equilibrium maximizes the total surplus.

# A Bread Cartel

One way to see this: suppose the bakeries form a **cartel**. They agree to charge \$6 and to bake only what buyers want at that price, 100 loaves.

- Buyers: CS $= \frac{1}{2} \times 100 \times 2 = 100$.
- Bakeries:
  PS $= 3 \times 100 + \frac{1}{2} \times 100 \times 1 = 350$.
- The trades between 100 and 200 no longer happen: a deadweight loss of $\frac{1}{2} \times 100 \times 3 = 150$. Total surplus falls to \$450.
- A price held **below** \$4 would leave trades undone too.



## Is the Outcome Pareto Efficient?

- An allocation is **Pareto efficient** if no feasible change could make someone better off without making anyone worse off.
- Since all the gains from trade are exhausted, no change in how many loaves are baked, or in who bakes and who buys them, can make a buyer or bakery better off without hurting another.
- So the equilibrium allocation is Pareto efficient, provided nobody outside the market is affected by the trades. If baking were noisy or polluting, the neighbours' costs would have to count too.
- An **external effect** is a cost or benefit that one person's action imposes on others, which the person acting does not take into account.

## When Does This Result Hold?

The Pareto efficiency of a competitive equilibrium is often read as a strong argument for markets. It holds only under four conditions.

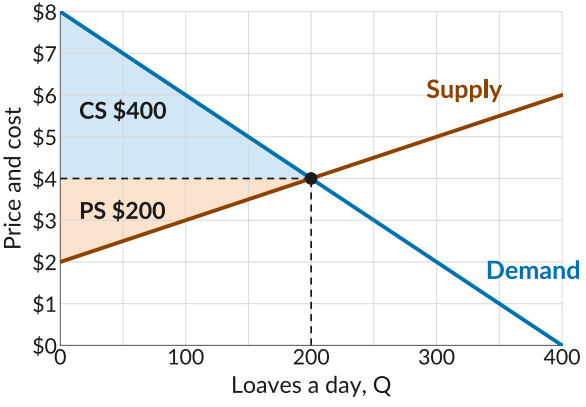
- Many buyers and sellers of identical goods.
- The market is at its equilibrium, with everyone a price-taker.
- Trade in the market has no external effects.
- Each trade is governed by a **complete contract**: one that covers everything either side cares about and can be enforced at almost no cost.

## In Practice

- Most firms sell goods that are differentiated in some way, as we saw before the midterm. Even bakeries selling identical loaves differ in location, service, and what else they sell.
- Evidence of price-taking is hard to find.
- Many goods have external effects, such as the carbon emissions from producing them.
- Contracts are sometimes incomplete. The buyer of a used car cannot tell whether it is reliable.

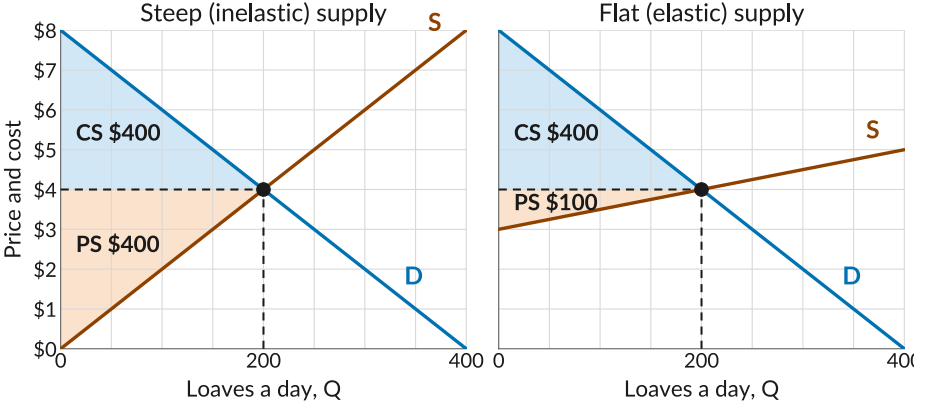
# Who Gets the Gains?

- Efficiency is one criterion for judging an allocation. Fairness is another.
- Surpluses say nothing about fairness, but they do show how the gains are split.
- Here buyers get 2/3 and sellers 1/3 of the surplus, because demand is steeper than supply.

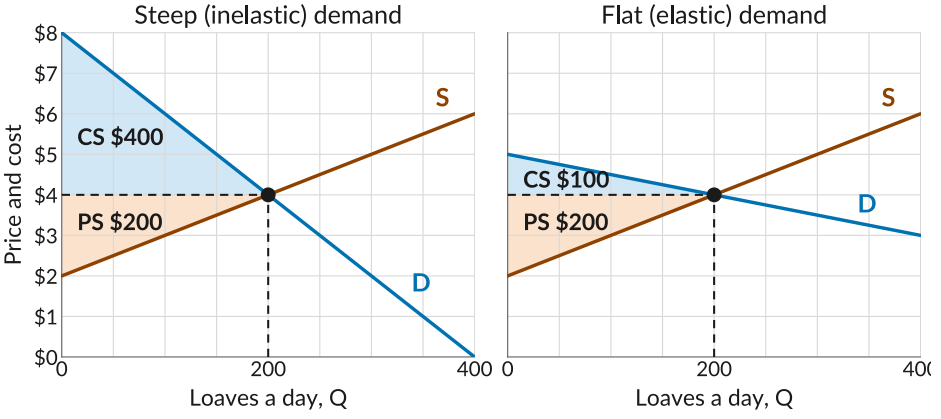


# Elasticity of Supply and the Split

The **elasticity of supply** measures how much the quantity supplied responds to the price. A flat supply curve is elastic, a steep one inelastic.



# Elasticity of Demand and the Split



## Elasticity and the Split

- A buyer's gain on a loaf is their willingness to pay minus the price, and a bakery's is the price minus marginal cost. Each side's surplus adds up these gaps.
- When demand is inelastic, many buyers value bread well above $4 and would pay far more than they do, so buyers gain a lot. When it is elastic, most buyers would pay only a little more than $4, and would walk away if it rose, so they gain little.
- The same goes for supply. Inelastic supply means many loaves cost well below $4 to bake; elastic supply means most loaves cost nearly $4.
- So how the gains are split depends on the relative elasticities of demand and supply: the less elastic side keeps more.